

Ratiq Ahamed MR

(Aspiring AI & Software Developer Intern)

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EDUCATION

EASWARI ENGINEERING COLLEGE

Bachelor of Technology

Major in Artificial Intelligence and Data science

Cumulative GPA: 7.79 %

Chennai, TN

2022-Present

WORK EXPERIENCE

ISMO BIO PHOTONICS, IITM (~15 employee Bio-Photonics Tools Manufacturer startup)

Machine Learning Engineer

IITMR, Chennai

July 2024 – August 2024

CORIZO PVT LTD

Data Science Intern

Remote

March 2024 – May 2024

PRODIGY INFOTECH

Data Science Intern

Remote

Feb 2024 – March 2024

PROFESSIONAL WORKS

1. Modern Website and Embedded RAG Chatbot for Ultrakote Printing and Packaging [ultrakote.in]:

- Designed and developed a responsive, modern website showcasing Ultrakote's comprehensive printing and packaging services.
- Integrated a Retrieval-Augmented Generation (RAG) chatbot capable of answering diverse customer queries accurately while incorporating censorship mechanisms using prompt engineering.
- Status: **Currently in use**

2. Domain-Specific Fine-Tuned Model for ISMO Biophotonics:

- Fine-tuned a Large Language Model to provide accurate and comprehensive responses regarding ISMO Biophotonics and the field of biophotonics.
- Status: **Currently in use**

3. Customer Care Sequential Chatbot for ISMO Biophotonics:

- Built a sequential chatbot with a modern user interface to streamline customer interactions, used methods like NLTK and Constructed a simple yet efficient ANN.
- Status: **Currently in use**

4. Question Paper Generator for Easwari Engineering College:

- Developed an application that takes question banks for each subject and generates mid-difficulty question papers in the exact required format. The app is capable of creating question papers for the entire college within a few minutes. And this app is being used by the college to create exam question paper for past 6 months.
- Status: **Currently in use**

5. Texture Generation GAN for gaming:

- A DC-GAN model trained for generating texture in gaming platform. Made this for a London based company which develops Roblox games.
- Status: **Currently in use**

SKILLS

Programming Languages: Python, C, HTML, CSS, JavaScript

Frameworks and Libraries: LangChain, LangGraph, LangSmith, Streamlit, CrewAI, Hugging Face, Groq, Pandas, Numpy, Matplotlib, Sklearn, Keras, Librosa, gTTS, FAISS, Transformers, Reactjs, TailwindCSS, PyTorch

Tools & Platforms: WordPress, Figma, Tableau

Database and ORM: SQLite, Neo4j, ChromaDB, MongoDB

Query Languages: SQL, Cypher

API: FastAPI, RESTful API

Prompt Engineering: Techniques: Chain of Thought, Few-Shot Learning, Zero-Shot Learning, Instruction Tuning.

Chatbots: Sequential, RAG, Agentic, Domain-Specific Chatbots creation and Integration.

Gen AI: Fine-Tuning (TTT,T2I), Prompt Tuning, Agents, LoRA, QLoRA

Cloud and Hosting: Web Hosting(cPanel), AWS

PERSONEL PROJECTS

VIRTUAL SALES PERSON.

- Developed an innovative virtual sales person companion that automatically triggers a pop-up interaction when customers add items to their cart. The LLM-powered assistant engages customers in natural conversation, creating an experience that mirrors the warmth and attentiveness of an in-store sales representative.
- Designed an intelligent recommendation system that helps customers find their perfect match by comparing their current selection with higher-end alternatives. The system analyzes factors like maintenance costs, rider's height, and personal preferences from its RAG database to suggest more suitable bikes, cleverly boosting both customer satisfaction and company revenue through premium product recommendations. And mainly it mimics a real sales person virtually.
- **Stack Used: MongoDB, Reactjs, FastAPI, GroqAPI, RAG**

INSTRUCTION PAIR DATASET CREATOR

- Created an instruction pair dataset creator that takes a text file from the user and converts it into instruction pair format. This tool uses NLP for proper chunking and LLM for creation.
- This project is novel in its approach to leveraging NLP and LLM technologies to automate the creation of instruction pairs, enhancing efficiency and accuracy.

BRAILLE CHATBOT:

- Developed a user interface chatbot designed for blind and deaf people. Users can communicate with this chatbot using a Braille keyboard and a microphone feature that allows them to talk to the chatbot. Additionally, the chatbot features a talking avatar that reads all messages aloud to the user.
- **Stack Used: Librosa, gTTS, Streamlit, Gemini API**

CERTIFICATIONS/ACHIEVEMENTS

- PYTHON COURSE (3/06/2022-26/12/2022) Completed a advance python course on LIVEWIRE
- ORACLE Certified Foundation Associate
- MongoDB CURD operations.
- Gen AI Study Jam - Vertex AI
- Analysis Case Study – IIT PALS
- Python Essentials – CISCO
- Introduction to IOT – NPTEL
- Ventura Hackathon – SSN (Participation Certificate)
- Assistant Sponsorship Head Of The Department (Easwari Engineering College 2023-2024)
- Secretary For Science Club(Easwari Engineering College 2022-2023)

ADDITIONAL

Languages: Fluent in English, Hindi, Urdu, Tamil

Non-Technical Skills: Offset printing